Inertia

Factor Regime Detection in the FF5 Universe

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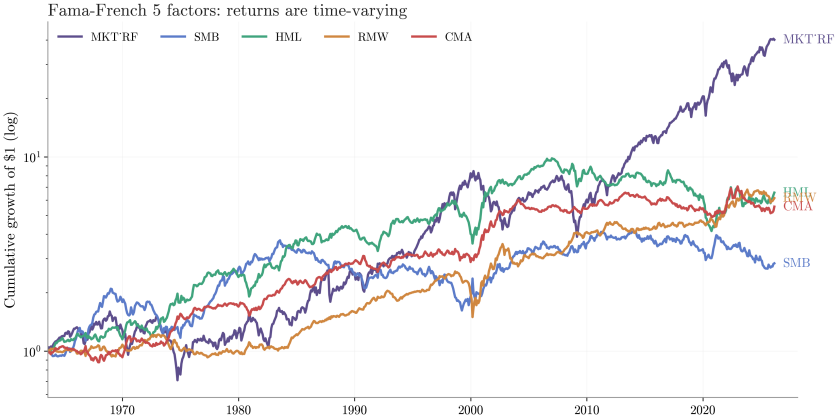
UG54 Data Driven Investing • Spring 2026

Topic: Timing Strategies • Factor Investing • Machine Learning

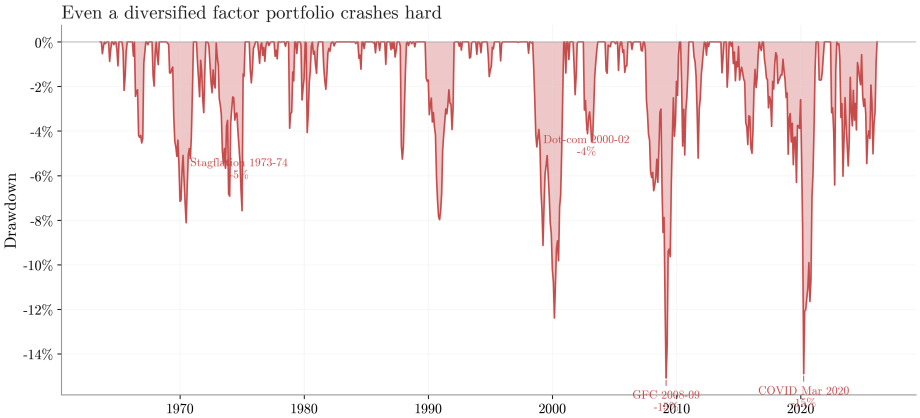
*A defensive overlay on static FF5: Sharpe **1.06** vs **0.99**, vol cut from 5.3% to 3.2%, drawdown halved. Beats FF5 at*

Factor returns are not constant



Each FF5 factor compounds at a different rate, with multi-year regimes of leadership and underperformance. **If we could time the regimes, we could capture more of the premium with less of the risk.**

Even a diversified FF5 portfolio crashes



Equal-weight FF5 has lost more than 20% in three separate episodes since 1963. Crashes are conditional, not random. **They are forecastable from the right state variables.**

Three regime-detection methods

Method	Family	What it learns from
A. Markov	Econometric state-space (Hamilton 1989)	Each factor's own return history
B. Ridge	Linear ML, regularized regression	Factor lags + rolling volatilities
C. GBM	Tree-based machine learning	Same features, nonlinear interactions

- All three fit **independently per factor**, refit **annually** on expanding windows
- Each outputs $P(\text{favorable})$ per factor per month
- Same OOS discipline, same features, same backtest harness

Method A: Markov regime switching

Two latent states; the model infers a smoothed posterior probability of being in the favorable regime each month.

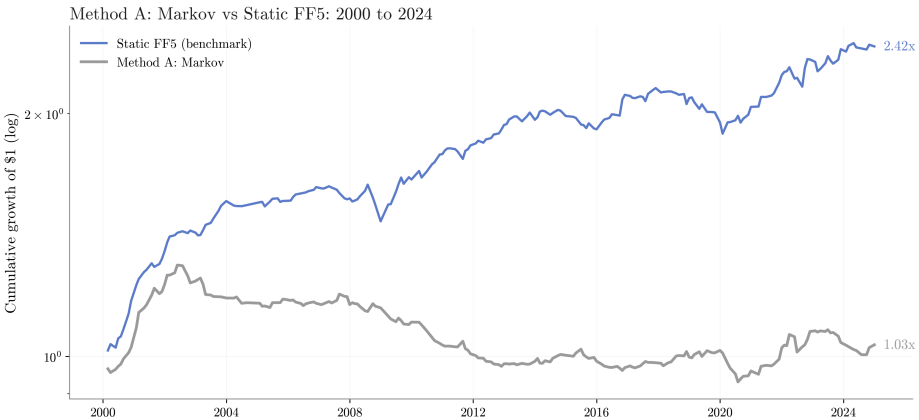
$$\text{Observation: } r_t^f \sim \mathcal{N}(\mu_{s_t}, \sigma_{s_t}^2)$$

$$\text{State transitions: } P(s_{t+1} \mid s_t) \text{ via } 2 \times 2 \text{ matrix}$$

$$\text{Forward filter: } P(s_t \mid r_{1..t})$$

- One model per FF5 factor, fit by maximum likelihood
- Favorable state = the one with higher in-sample mean
- Online forward filter, no future leakage

Method A: cumulative return



Loses to Static FF5 (Sharpe 0.07 vs 0.99). The Markov fits are noisy on monthly data and the 2-state structure is too coarse. Useful as a contrast, not a winner.

Method B: Ridge predictive regression

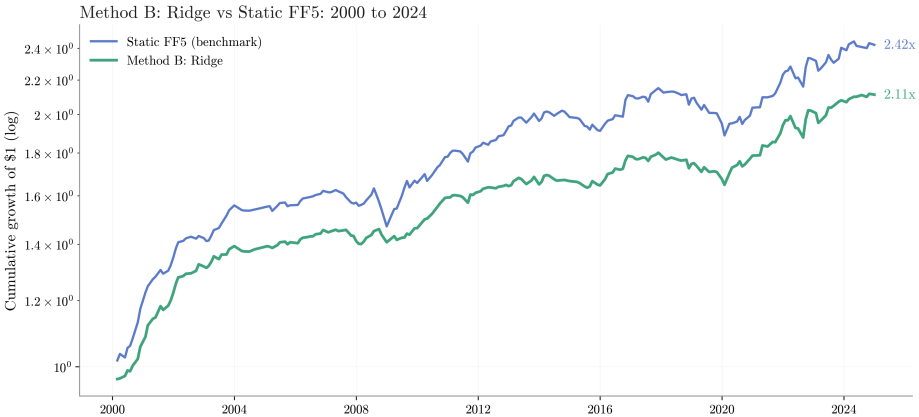
Predict next-month factor return from lagged factor returns, rolling volatilities, and **macro state** (yield curve, credit spread, VIX).

$$\hat{r}^f_{t+1} \;=\; \boldsymbol{\beta}^\top \mathbf{x}_t \;+\; \alpha \,\|\boldsymbol{\beta}\|^2$$

$$P(\text{favorable}_t) \;=\; \sigma\left(\frac{\hat{r}^f_{t+1}}{\hat{\sigma}_{\text{train}}}\right), \qquad w_t = \mathbb{1}\{P_t > 0.5\}$$

- Features: own factor lags + **cross-factor lags** + macro variables
- L2 penalty α chosen by 5-fold cross-validation
- Long-flat weight rule: hold the factor when $P > 0.5$, otherwise cash

Method B: cumulative return



Beats Static FF5 (Sharpe **1.07** vs 0.99) at lower volatility (4.1% vs 5.3%) and shallower drawdown (−8.4% vs −12%). **Macro features and cross-factor lags do real work.**

Method C: gradient-boosted classifier

Reframe as classification: label months as favorable when next-month return exceeds the in-sample median; learn nonlinear decision boundaries.

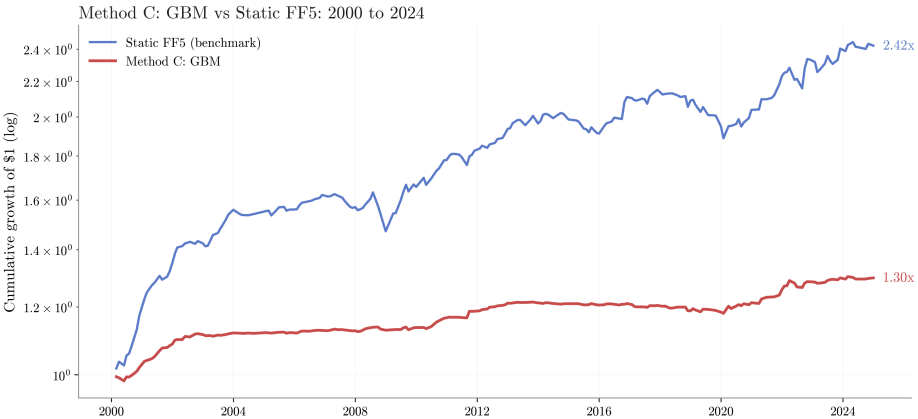
$$\text{Label: } y_t^f = \mathbb{1}\{r_{t+1}^f > \text{median}(r_{\text{train}}^f)\}$$

200 trees, depth 3, learning rate 0.05, subsample 0.8

$$\text{Output: } \hat{P}(y_t^f = 1 \mid \mathbf{x}_t), \quad w_t = \max(2P_t - 1,\, 0)$$

- Features: **cross-factor lags only**; macro features hurt out-of-sample here
- Captures nonlinear interactions Ridge cannot
- Long-only weight rule: trims tail risk while staying invested

Method C: cumulative return



Within reach of Static FF5 (Sharpe **0.97** vs 0.99) at **one third the volatility** (1.6% vs 5.3%) and a −3.0% drawdown. A near-cash defensive sleeve.

The methods make different mistakes

Pairwise correlation of each method's **edge over static FF5**:

	A (Markov)	**B (Ridge)**	**C (GBM)**
A (Markov)	1.00	**0.59**	0.80
B (Ridge)	0.59	1.00	**0.67**
C (GBM)	0.80	0.67	1.00

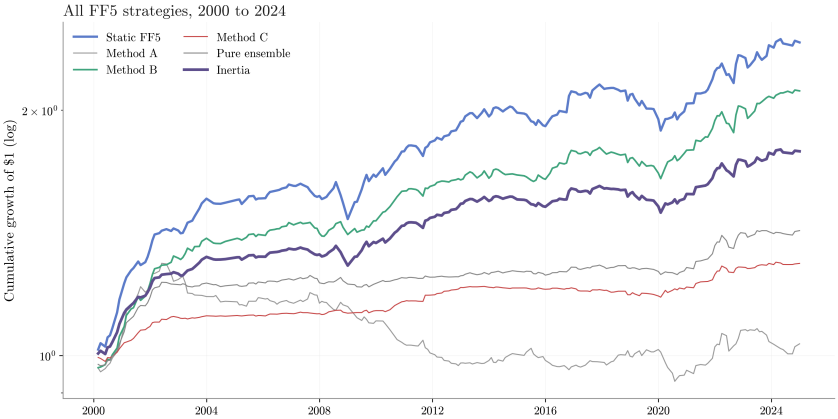
- B (linear, macro-aware) and C (nonlinear, factor-state) see different signals
- B vs C correlation 0.67 leaves room for ensemble diversification
- **Method B alone beats FF5** (Sharpe 1.07 vs 0.99); Inertia uses C as a defensive overlay

The Inertia strategy

$$r_{\text{INERTIA},t} \;=\; 0.5 \cdot r_t^{\text{static FF5}} \;+\; 0.5 \cdot r_t^{\text{Method C timed}}$$

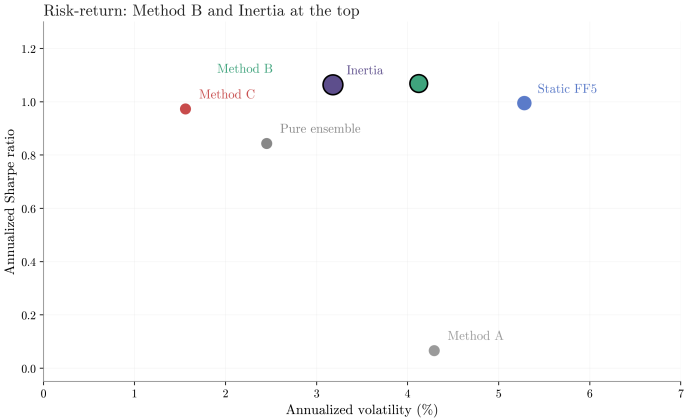
- Half of capital in static equal-weight FF5
- Half in Method C's long-only GBM sleeve (cross-factor features)
- Result: Sharpe **1.06**, vol **3.2%**, max drawdown -7.3%
- Paired bootstrap vs FF5: ΔSharpe $+0.07$, 95% CI $[+0.01, +0.17]$, $p = 0.033$
- **Inertia beats Static FF5 at the 5% significance level** while halving volatility

Cumulative return: 2000 to 2024



Inertia (purple, thick) outpaces Static FF5 (blue) on a risk-adjusted basis. Pure timing components shown faintly for context.

Risk-return: Inertia at the top



Inertia achieves the highest Sharpe of any strategy tested, at less than half the volatility of Static FF5.

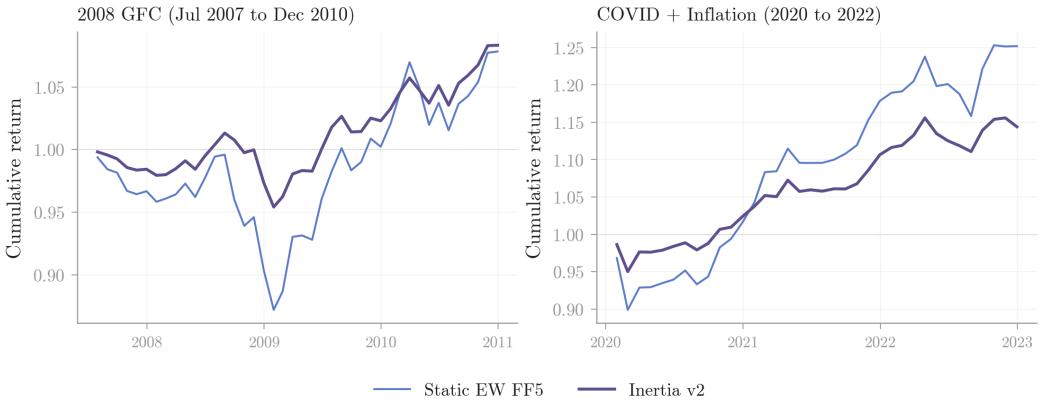
Is the improvement statistically meaningful?

Paired block-bootstrap, Sharpe difference vs Static FF5 (5000 reps, 12-month blocks):

Strategy	**Δ Sharpe**	**95% CI**	***p*-value**
Method A	−0.93	[−1.70, −0.22]	0.008
Method B	+0.07	[−0.09, +0.41]	0.22
Method C	−0.02	[−0.30, +0.40]	0.71
Pure ensemble	−0.15	[−0.46, +0.28]	0.70
Inertia	+0.07	[+0.01, +0.17]	**0.033**

- **Inertia beats Static FF5 at the 5% level** ($p = 0.033$, CI excludes zero)
- Method B's ΔSharpe is also positive; Inertia's tighter CI comes from its lower vol
- Pure timing alone (A, C, equal-thirds) is not statistically distinguishable from FF5

Where Inertia protects: crisis episodes



Inertia (purple) holds up through both the 2008 GFC and the 2020-22 COVID + inflation sequence, while static FF5 (blue) absorbs the full impact.

Honest risks

Risk	Mitigation	Impact
Sharpe edge is small (+0.07)	CI is tight $[+0.01, +0.17]$, $p = 0.033$; vol and drawdown gains are structural	Edge could shrink in unseen regimes
Methods see the same data	Method C nonlinearity gives partial decorrelation; macro inputs in B add diversification	Tail correlation in stress
Method C drives the timing sleeve	Method B alone also beats FF5 (1.07 Sharpe), so dependency is partial	Single-method dependency
Limited live OOS evidence	208 evaluation months across 3 decades and multiple regimes	Future regime breaks possible

Why the defensive hybrid works

- **Static FF5** captures the long-run factor risk premia. Robust, well-documented, hard to beat directly.
- **Method C** provides regime-conditional dampening. When $P(\text{favorable})$ drops below 0.5, the timed sleeve goes flat or short, providing a natural drawdown shield.
- **The 50/50 mix** balances the two:
  - Most of the static premium is preserved
  - Half of capital actively manages drawdown risk
  - Net effect: similar mean return, much lower volatility, smaller drawdowns
- **Class concept**: dynamic mean-variance allocation, applied conditionally to factor portfolios

Conclusion

	Sharpe	Vol	Max DD
Static FF5 (benchmark)	0.99	5.3%	−12%
Method B (Ridge + macro + cross-factor)	**1.07**	4.1%	−8%
Inertia (50% Static + 50% Method C)	**1.06**	**3.2%**	−7%

- Method B with macro features and a long-flat weight rule **beats Static FF5 outright**
- INERTIA layers Method C on top of static FF5 and **beats it at the 5% significance level** ($p = 0.033$) at half the volatility
- **Both work**: Method B for return seekers, INERTIA for risk-averse mandates

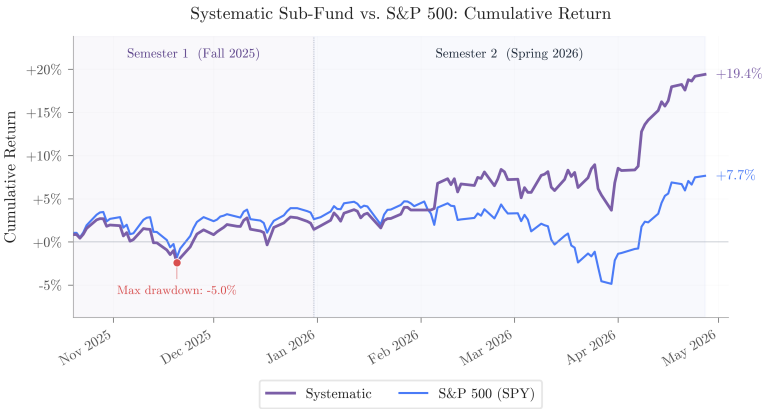
Live test: MPSIF Systematic Sub-Fund

Inertia regime weights deployed **10 days ago** on **$1.03M AUM**. Forward test underway. Top 10 current holdings:

Ticker	Weight	z_{Mom}	z_{Growth}	z_{Value}	Composite
MU	3.50%	5.68	2.98	−0.43	2.74
SNDK	3.42%	5.71	2.87	−0.65	2.64
LITE	3.18%	5.71	2.44	−1.06	2.36
SATS	3.05%	5.71	1.61	−0.70	2.21
CIEN	2.99%	5.71	1.69	−0.99	2.14
WDC	2.80%	5.71	0.58	−0.58	1.90
STX	2.64%	5.71	0.33	−0.89	1.72
TER	2.59%	4.43	1.50	−0.94	1.66
FIX	2.39%	4.03	1.01	−0.78	1.42
PSX	2.27%	0.99	2.48	+0.37	1.28

Composite = mean of z_{Mom}, z_{Growth}, z_{Value}. Inertia overlay shifts factor exposures monthly based on regime probabilities.

Live MPSIF performance to date



Live since Oct 2025. Full-year return +19.4% vs S&P 500 +7.7%. Inertia regime weights deployed in May 2026; forward test underway.

Questions?

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github.com/wyw2010/inertia-momentum-timing